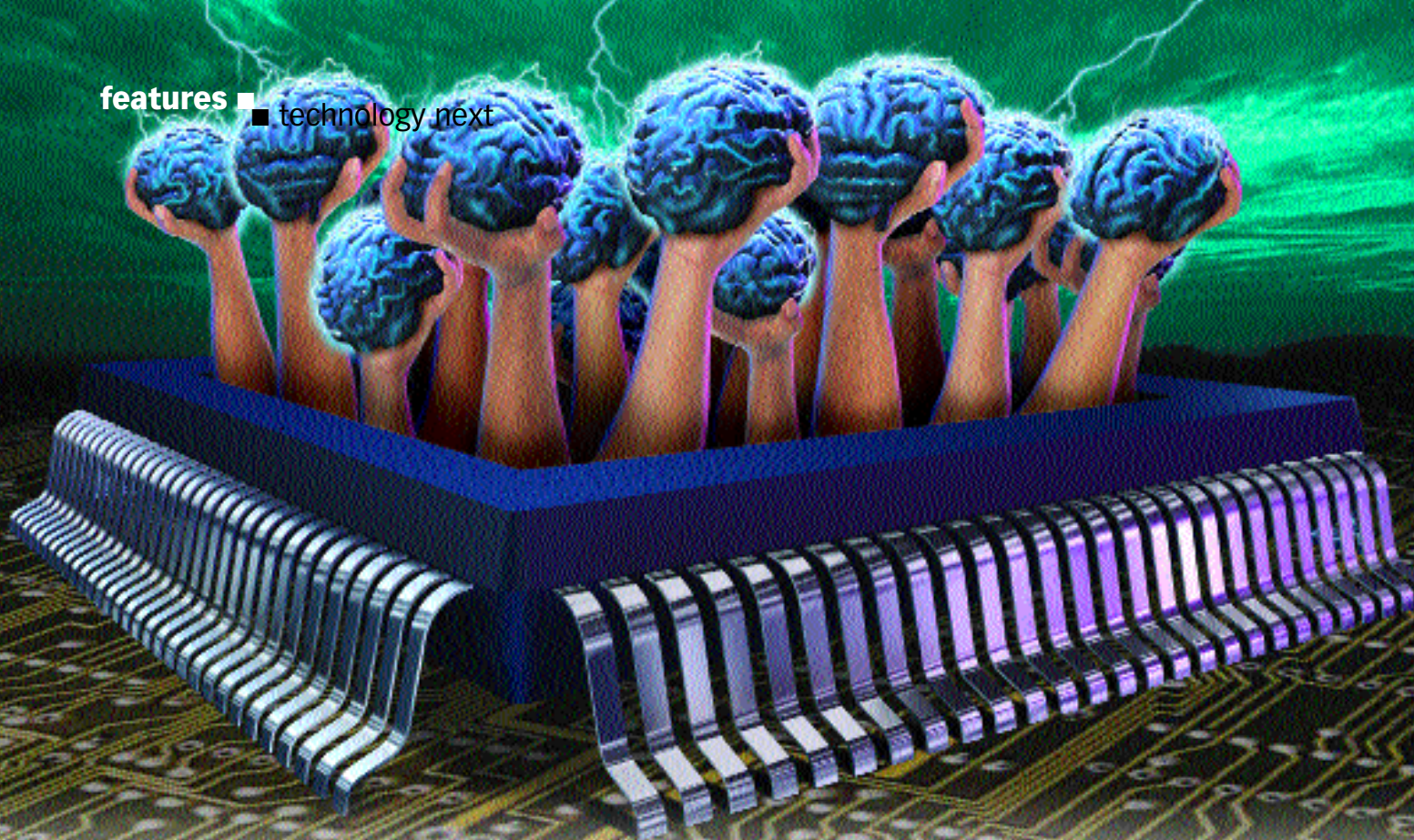




GRAPHIC DESIGN: Ram

Memory Mat-

Computing at the speed of thought: a look at the next step in the evolution of cheap, fast and dense solid-state memory

First there was the race for greater density, then came the quest for newer technology, heralding an accelerated adaptation of RAM. This almost quick-time implementation of memory catalysed growth in other areas of technology, rushing us to the age of nanosecond computing.

If it were not for these advancements, tiny cellular devices with 500-address memories, PDAs that let you watch movies and MP3 players that store your entire music collection would never have seen the light of day. In more familiar territory, even computer memory has seen numerous architectural iterations—from EDO RAM to the numerous flavours of SDRAM, all the way to RDRAM and DDR.

The need for speed

In the continued quest for the perfect memory, companies are struggling to

achieve the golden balance between factors such as data transfer speed, retention, density and low costs. However, each implementation of RAM has greater strength in some of these areas than others. Today's computers are composed of different types of data storage—the hard disk that stores the permanent data, the RAM (either SDRAM, DDR or RDRAM), which is used to store programs that are loaded and running, and finally, SRAM that is used in high-speed implementations such as the cache on processors.

RAM (SDRAM, DDR and RDRAM) as we know it today, can not retain data once the power has been switched off—this is a definite shortcoming. SRAM is significantly faster than other forms of RAM but is expensive to produce on a very large scale, which is why it is used only in small quantities in computer sys-

tems. Though manufacturers have figured a workaround to the problem of data loss in case of Flash RAM, this kind of RAM has an associated life cycle before it begins to lose its ability to retain data.

While the architecture of each solid-state storage technology differs, they are still fabricated using the same fundamental design. This is the problem that is throttling the power and functionality in today's computing applications—be it on the desktop or on mobile devices, and this limitation is the main driver of the innovations in fabricating newer forms of volatile storage.

The search begins

Considering the overall size of the memory chip market (\$25 billion, (approx Rs 1,25,000 crore) in 2001, even in the face of recession and sagging economies), it is of little surprise that some of the biggest →